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1. 习题三:

20.

$$(1) f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^{\frac{x}{2}} \frac{3}{4} x dy = \frac{3}{8} x^2, & 0 \leq x \leq 2 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_{2y}^2 f(x, y) dx = \frac{3}{2} - \frac{3}{2} y^2, & 0 \leq y \leq 1 \\ 0, & \text{others} \end{cases}$$

$$(2) \because f(x, y) \neq f_X(x) f_Y(y), 0 \leq x \leq 2, 0 \leq y \leq \frac{x}{2}$$

$\therefore X, Y$ 不相互独立

22.

(1) 设甲乙分别在 9 时 X 分和 9 时 Y 分到达车站, 则由题意得

$$f_X(x) = \begin{cases} \frac{1}{30}, & 0 \leq x \leq 30 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{40}, & 10 \leq y \leq 50 \\ 0, & \text{others} \end{cases}$$

$\therefore X, Y$ 相互独立

$$\therefore f(x, y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{1200}, & 0 \leq x \leq 30, 10 \leq y \leq 50 \\ 0, & \text{others} \end{cases}$$

$$P(X < Y) = 1 - \int_{10}^{30} \int_{10}^x f(x, y) dy dx = \frac{5}{6}$$

故甲先到的概率为 $\frac{5}{6}$

(2) 设 A = “先到者等后到者时间不超过 10min”

$$P(A) = P(|X - Y| \leq 10) = P(X - 10 \leq Y \leq X + 10) = \int_{10}^{20} \int_{10}^{x+10} f(x, y) dy dx + \int_{20}^{30} \int_{x-10}^{x+10} f(x, y) dy dx = \frac{1}{3}$$

故先到者等后到者时间不超过 10min 的概率为 $\frac{1}{3}$

(1) 设周收入为 Z

$$P(Z \leq z) = \int_{0.1}^{0.2} P(X = x, Y = \frac{z}{x}) dx = \int_{0.1}^{0.2} f(x, \frac{z}{x}) dx = 1 - e^{-z}, z > 0$$

$$\therefore F_Z(z) = \begin{cases} 1 - e^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

$$f_Z(z) = \begin{cases} e^{-z}, & z > 0 \\ 0, & z \leq 0 \end{cases} \quad (2) f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^{+\infty} f(x, y) dy = 10, & 0.1 < x < 0.2 \\ 0, & \text{others} \end{cases}$$

$$(3) f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} xe^{-xy}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$\therefore P(Y \geq 1|X = 0.15) = \int_1^{+\infty} f(y|0.15) dy = e^{-0.15}$$

$$P(Y \geq 1|X = 0.2) = \int_1^{+\infty} f(y|0.2) dy = e^{-0.2}$$

2. 3.2-6 补充题一:

$$f(x, y) = f_X(x)f_{Y|X}(y|x) = \begin{cases} 8xy, & 0 \leq x \leq 1, x \leq y \leq 1 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^y 8xy dx = 4y^3, & 0 \leq y \leq 1 \\ 0, & \text{others} \end{cases}$$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2x}{y^2}, 0 \leq y \leq 1, 0 \leq x \leq y$$

$$P(X > \frac{1}{2}|Y = \frac{2}{3}) = \int_{\frac{1}{2}}^{\frac{2}{3}} f_{X|Y}(X > \frac{1}{2}|Y = \frac{2}{3}) dx = \int_{\frac{1}{2}}^{\frac{2}{3}} \frac{9}{2} x dx = \frac{7}{16}$$

3.2-6 补充题二:

$$(1) \int_{-\infty}^{+\infty} f(x, y) dy dx = \frac{c}{3} = 1$$

$$\therefore c = 3$$

$$P(X \leq Y) = \int_0^1 \int_0^y f(x, y) dx dy = 0.5$$

$$P(X \leq 0.3, X \leq Y) = P(X \leq Y) - P(X > 0.3, X \leq Y) = P(X \leq Y) - \int_{0.3}^1 \int_{0.3}^y f(x, y) dx dy = 0.0585$$

$$P(X \leq 0.3 | X \leq Y) = \frac{P(X \leq 0.3, X \leq Y)}{P(X \leq Y)} = 0.117$$

$$(2) F_X(x | X \leq Y) = P(X \leq x, X \leq Y) = \frac{P(X \leq x, X \leq Y)}{P(X \leq Y)} = \frac{\int_0^x \int_u^x f(u, v) dv du + \int_0^x \int_x^1 f(u, v) dv du}{P(X \leq Y)} = \frac{\frac{1}{2}(x^2 + x^3 - x^5)}{0.5} = x^2 + x^3 - x^5$$

3.3-6 补充题一:

(1) 由题意得

$$f(x, y) = \begin{cases} \frac{1}{2}, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{others} \end{cases}$$

$$P(U = 0, V = 0) = P(X \geq Y) = \int_0^1 \int_0^x f(x, y) dy dx = \frac{1}{4}$$

$$P(U = 0, V = 1) = 0$$

$$P(U = 1, V = 0) = P(X < Y \leq 2X) = \int_0^1 \int_x^{2x} f(x, y) dy dx = \frac{1}{4}$$

$$P(U = 1, V = 1) = P(Y > 2X) = \int_0^1 \int_{2x}^1 f(x, y) dy dx = \frac{1}{2}$$

故 U、V 的联合分布列为

p_{ij}		U		$p_{\cdot j} = \sum_i p_{ij}$
		0	1	
V	0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
	1	0	$\frac{1}{2}$	$\frac{1}{2}$
$p_{i \cdot} = \sum_j p_{ij}$		$\frac{1}{4}$	$\frac{3}{4}$	1

$$(2) \because P(U = 0, V = 1) \neq P(U = 0)P(V = 1)$$

$\therefore$ U、V 不相互独立

3.3-6 补充题二:

(1) 由题意得

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$P(U = k, V \leq x) = \begin{cases} P(k \leq X < k + x) = \int_k^{k+x} f(x) dx = e^{-k\lambda} - e^{-(k+x)\lambda}, & k \in Z, 0 \leq k \leq [\lambda], 0 \leq x \leq 1 \\ 0, & otherwise \end{cases}$$

$$(2) P(U = k) = P(k \leq X < k + 1) = \int_k^{k+1} f(x) dx = e^{-k\lambda} - e^{-(k+1)\lambda} = e^{-k\lambda}(1 - e^{-\lambda})$$

$$P(V \leq x) = \sum_{k=0}^{[\lambda]} P(U = k, V \leq x) = \frac{1 - e^{([\lambda]+1)\lambda}}{1 - e^{-\lambda}}(1 - e^{-x\lambda})$$

故 $P(U = k, V \leq x) \neq P(V \leq x)P(U = k)$, 故 U, V 相互独立